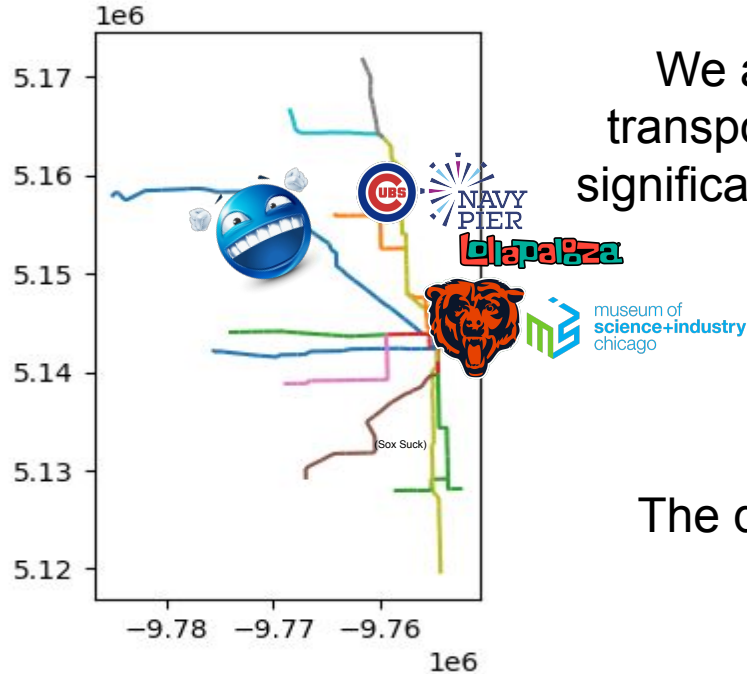


A Geospatial Approach to Chicago Public Transportation



Henry Rotunno

Investigating Transportation Equity



We all love Chicago for its comprehensive public transportation system. The L trains and buses make it significantly easier to travel around the city without a car.

However,

The quality and quantity of the infrastructure varies greatly depending on location.

Safety?

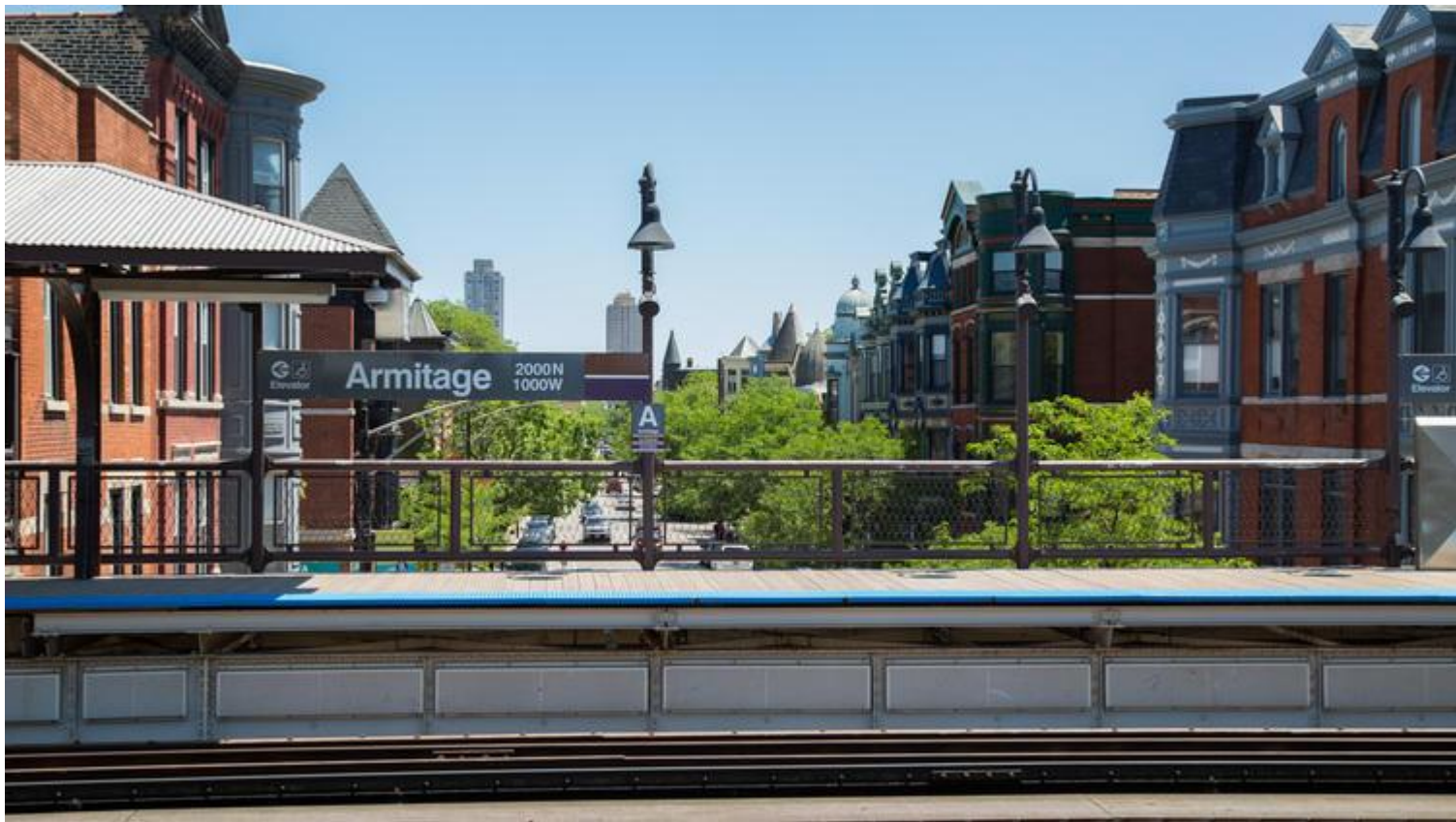


Cleanliness?

Redline stop at Monroe

Accessibility?

Design!
Accessibility!



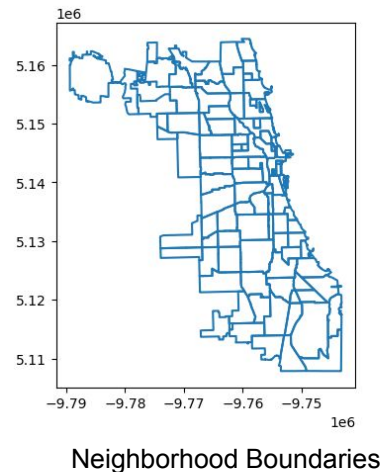
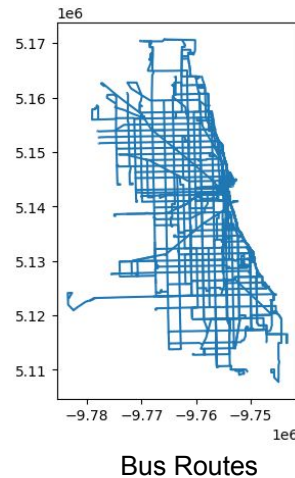
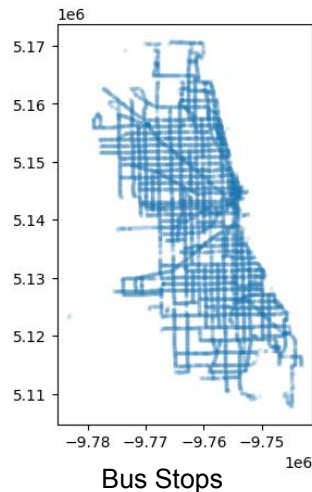
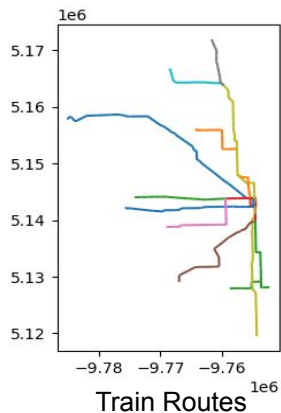
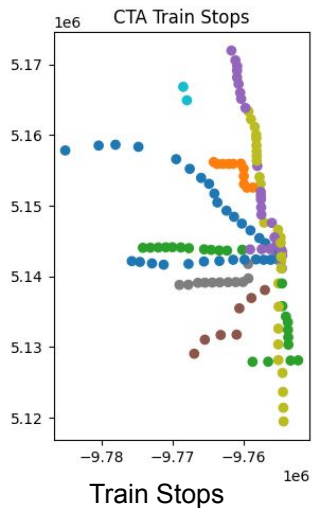
Problem: Despite the CTA's huge budget, there are **transit deserts** in the city. Notably, neighborhoods in the south and west contain fewer options and are overall worse quality.

Analysis: It is commonly known that Chicago's south and west neighborhoods consist of primarily marginalized communities. This is a product of redlining and racist housing policies. As a result, these neighborhoods receive minimal funding and attention. This leads to slow industry, a lack of jobs, poor infrastructure, instability, and crime.



Goal: I want to visualize demographic data in conjunction with transportation data to reveal transportation injustice in Chicago.

Datasets



GEOIDFQ	ST	Name	Latitude	Longitude	Square_Miles	Total_Households	Average_Household_Size	Percent_Single_Mothers	Percent_Ir
1400000US17031640700	IL	6407	41.776412	-87.751940	0.25	1063	3.13	NaN	
1400000US17031640100	IL	6401	41.781929	-87.739822	0.12	493	2.98	0.0	
1400000US17031980100	IL	9801	41.785971	-87.750892	1.15	0	NaN	NaN	
1400000US17031560700	IL	5607	41.796467	-87.757523	0.26	1319	2.91	0.0	
1400000US17031835200	IL	8352	41.796640	-87.745280	0.38	680	3.28	100.0	



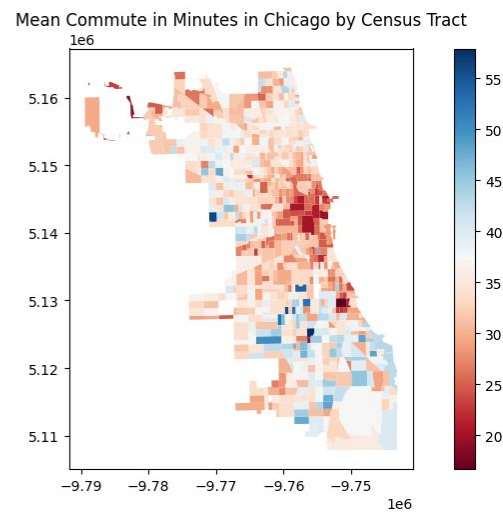
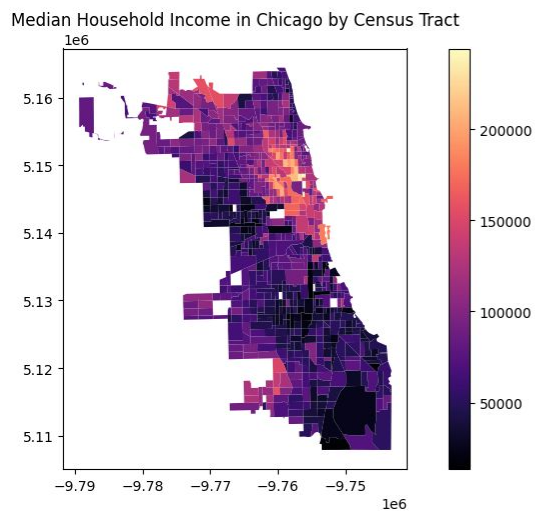
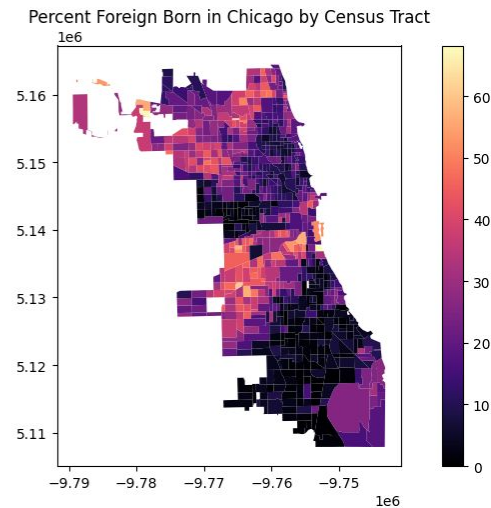
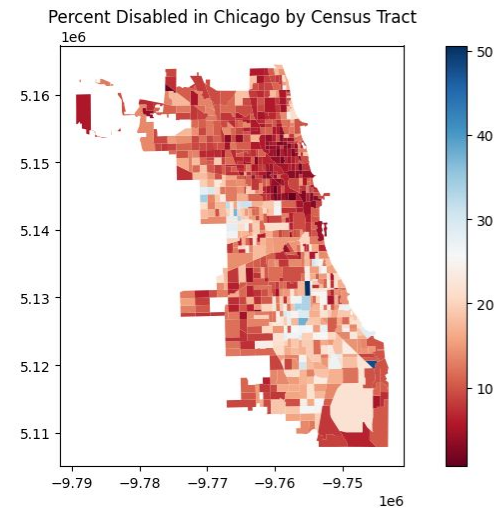
**CHICAGO
DATA PORTAL**

Chicago
Census
Tracts
(ACS)



**AMERICAN
COMMUNITY
SURVEY**
U.S. CENSUS BUREAU

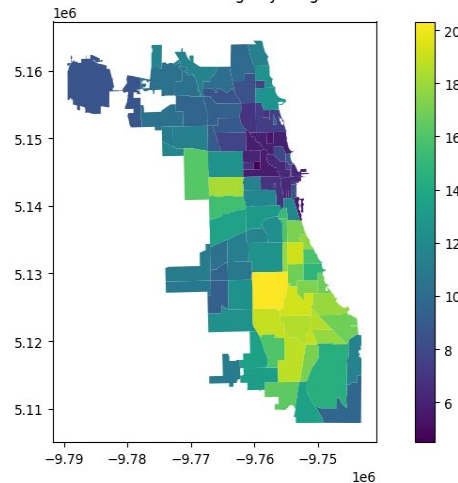
Demographic and Economic Data in Chicago



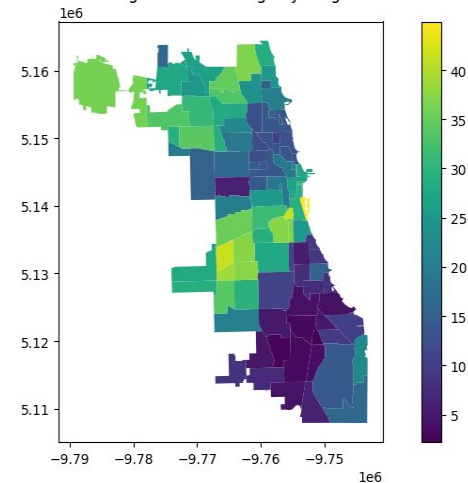
Geopandas
+ Matplotlib

Demographic and Economic Data in Chicago Neighborhoods

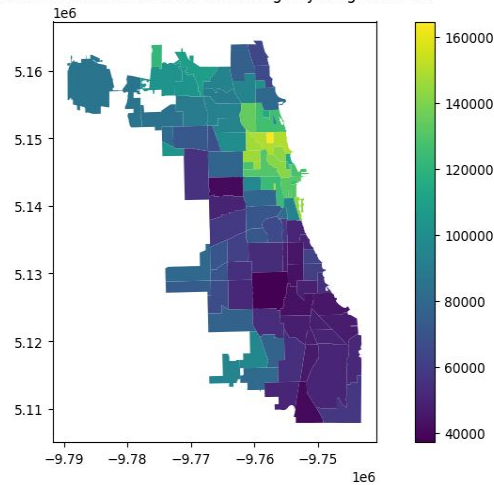
Percent Disabled in Chicago by Neighborhood



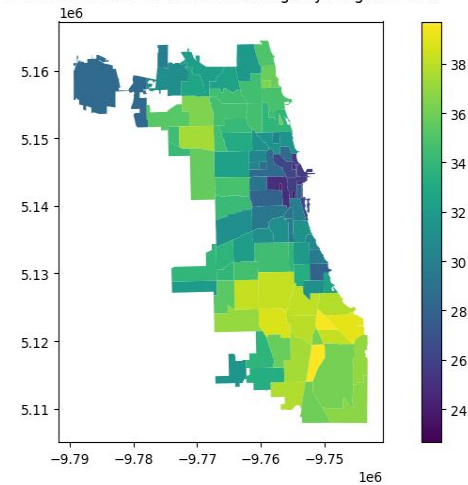
Percent Foreign Born in Chicago by Neighborhood



Median Household Income in Chicago by Neighborhood



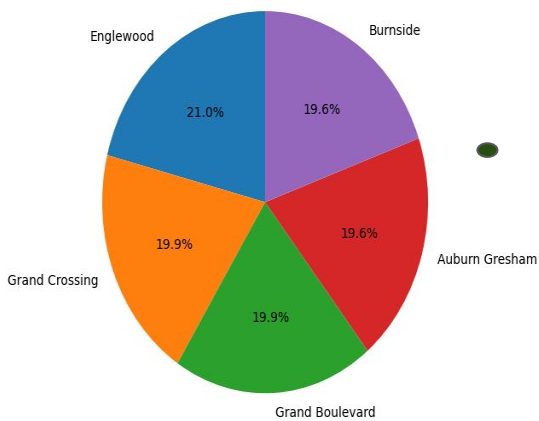
Mean Commute in Minutes in Chicago by Neighborhood



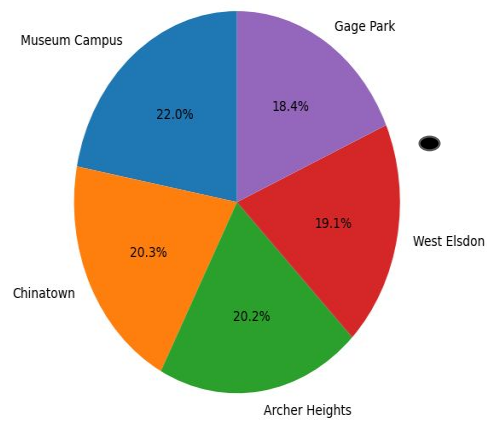
Spatial join +
aggregation

Demographic and Economic Data in Chicago Neighborhoods

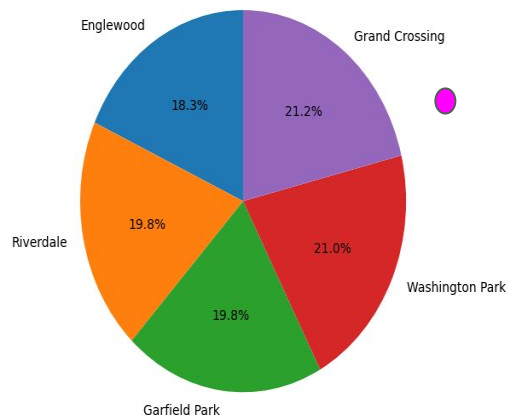
5 Highest Neighborhood Disability Rates



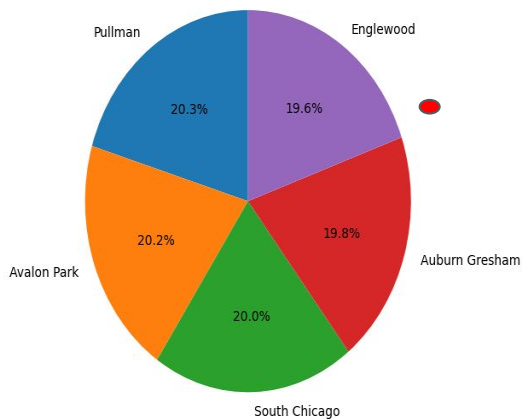
5 Highest Neighborhood Foreign Born Rates



5 Lowest Neighborhood MHHI

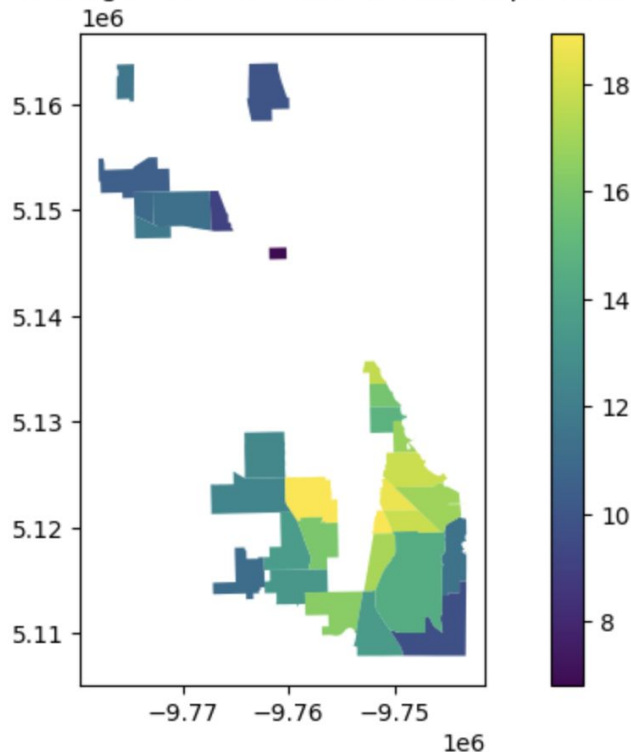


5 Highest Commutes Times

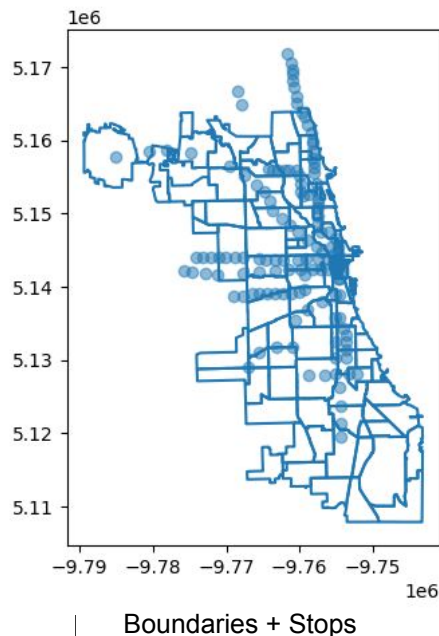


Transportation

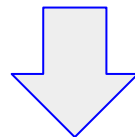
Disability Rate of Neighborhoods Without an L Stop Within 0.5 Miles



A walkable area is commonly described as within 0.5 miles.



30% of neighborhoods don't have a station within 0.5 miles!!!

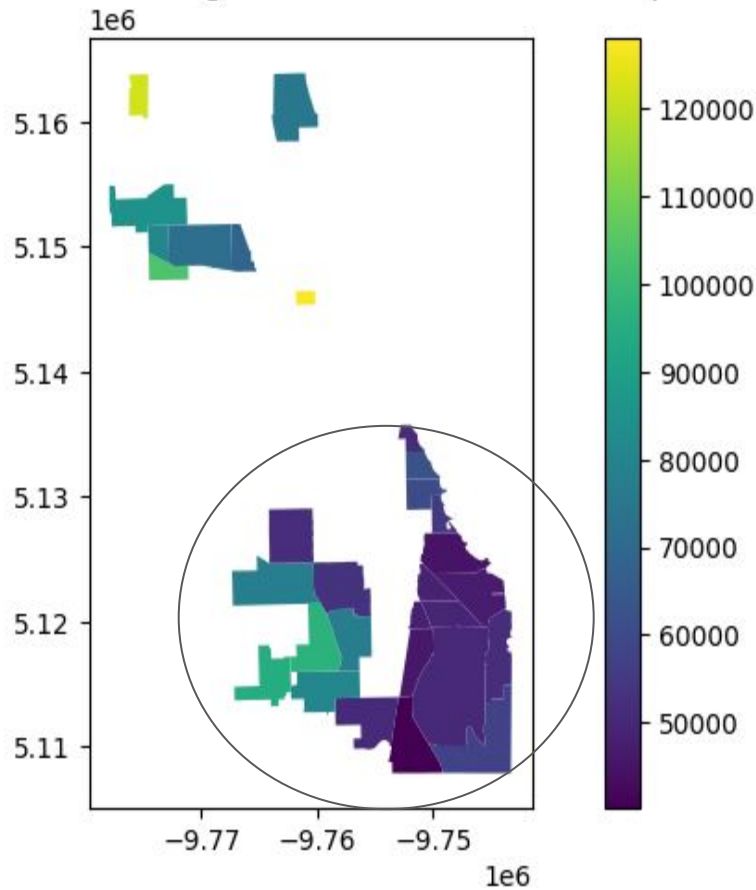


Can you imagine having to travel over half a mile in a wheelchair just to make it to a train?



These are the neighborhoods that **don't** have an L train stop within 0.5 miles symbolized by percent disability. Surprised?

Median Household Income of Neighborhoods Without an L Stop Within 0.5 Miles

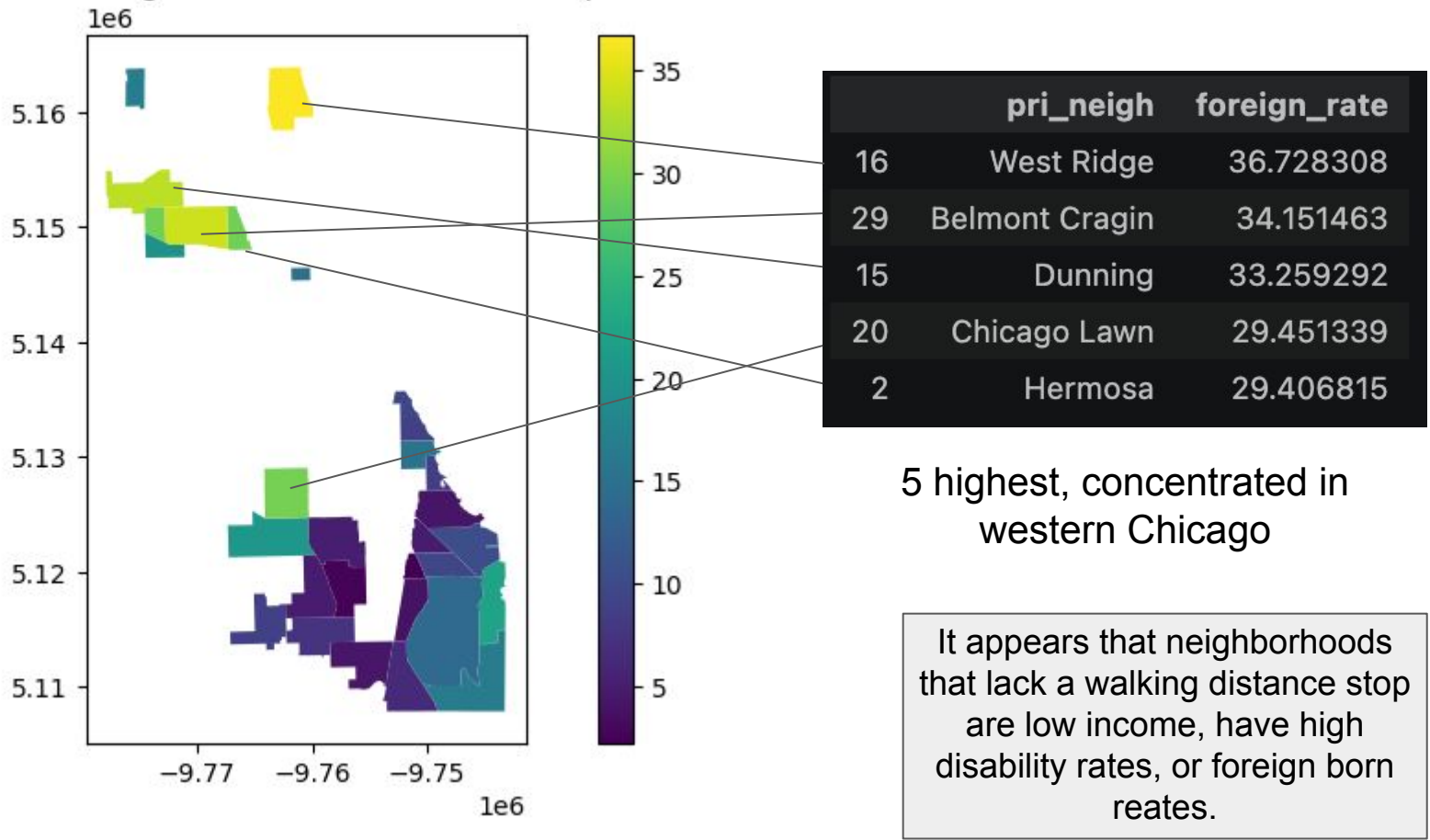


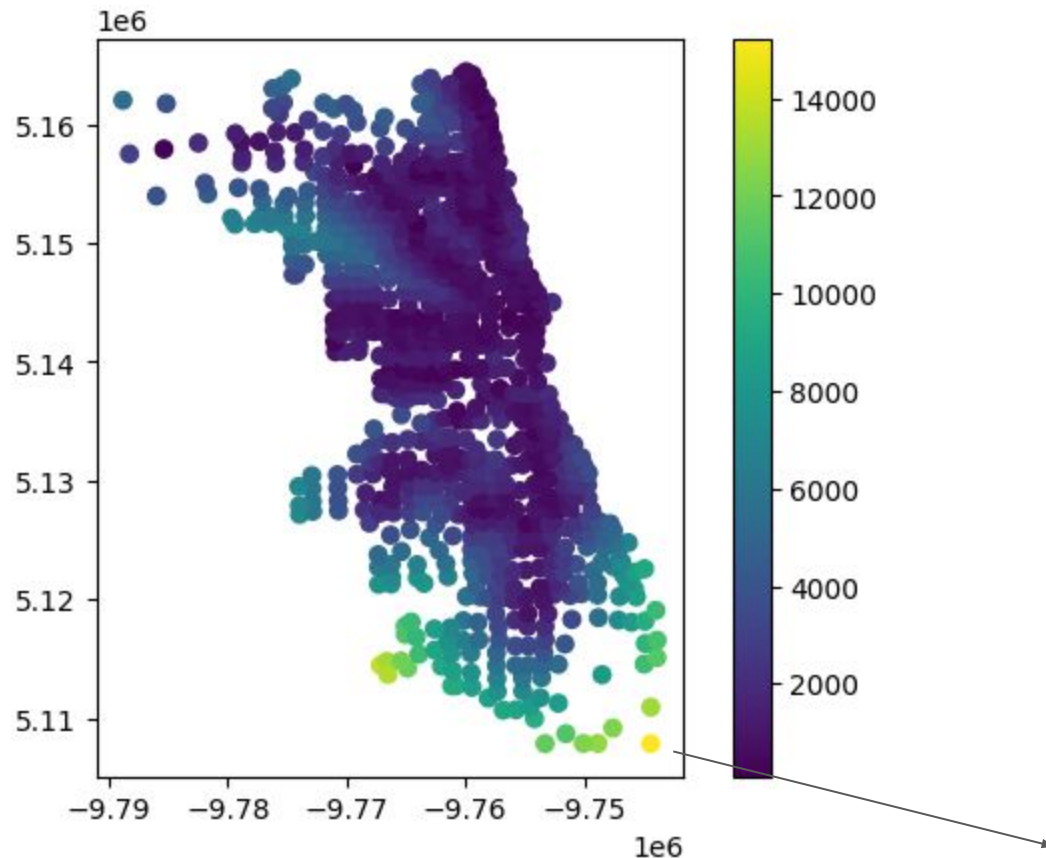
Less affluent areas
have worse or no
public transport
options in Chicago.

pri_neigh	Median_Household_Income
Riverdale	40219.083333
South Shore	44962.666667
Pullman	46003.750000
South Chicago	46997.750000
Burnside	47912.222222
Avalon Park	48506.000000
Calumet Heights	49826.357143

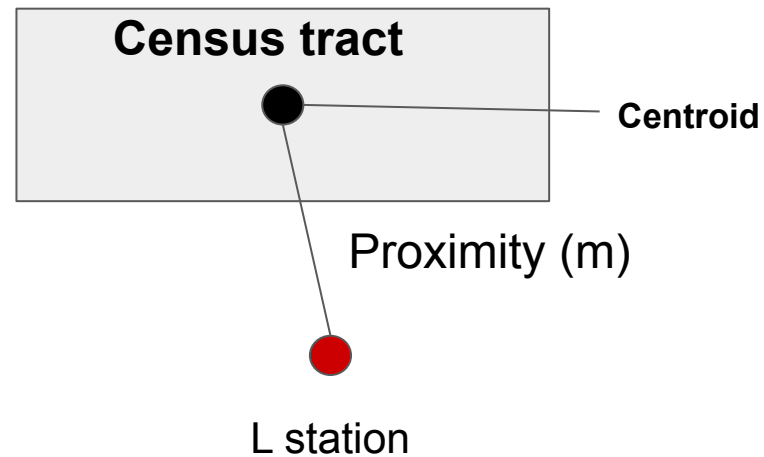
Bottom 7 neighborhoods

Foreign Born Rate for Neighborhoods Without an L Stop Within 0.5 Miles





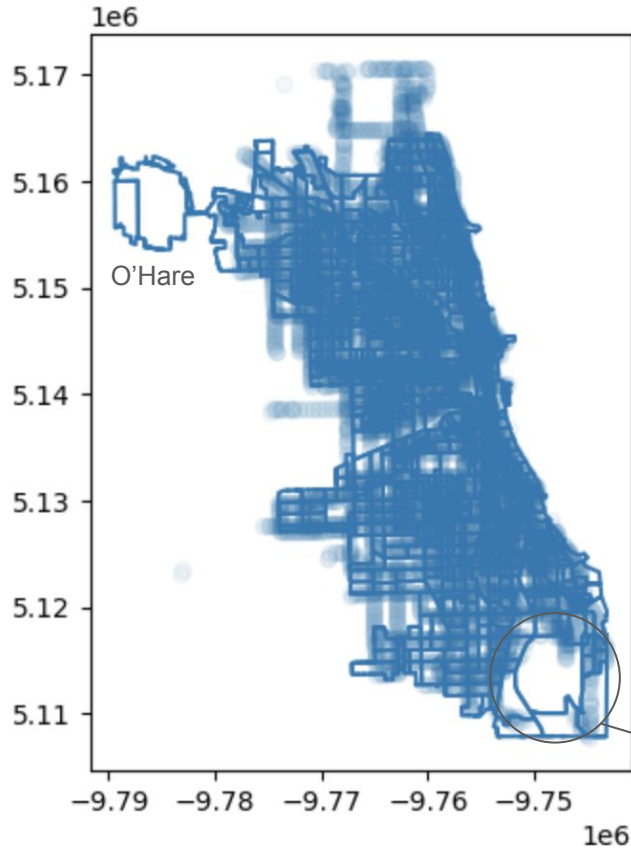
Proximity of census tracts (centroids) to the nearest station in meters



```
1 nearest['distance'].max()  
✓ 0.0s  
15239.721855129934
```



Bus Stops



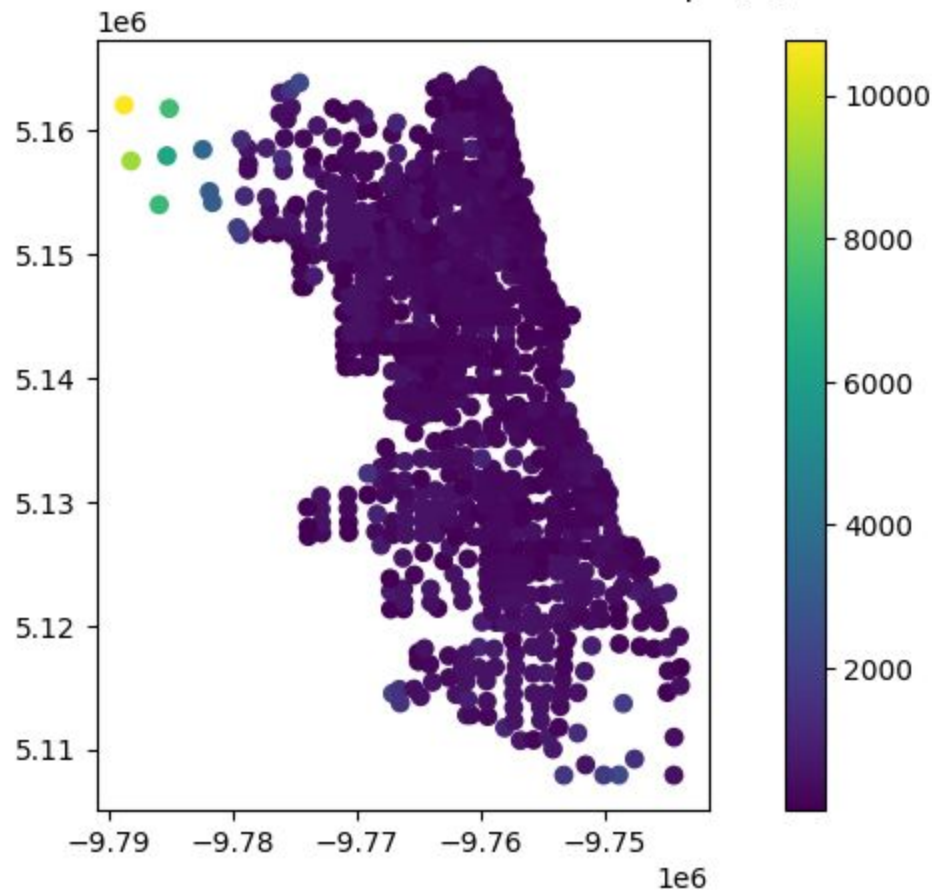
Census tracts and bus stops

Chicago's Bus System

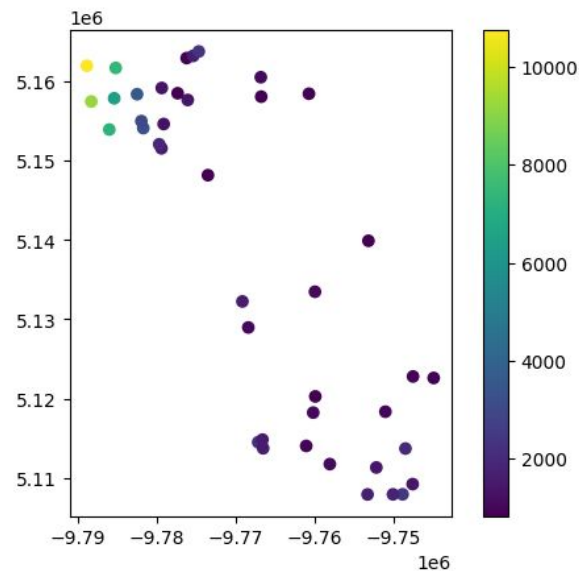
- + Significantly more stops
- + Generally safer / cleaner
- Typically slower
- Prevalence of ghost buses
- Some dead areas
- CTA funding diminishing



Distance of each Census Tract to Bus Stops (m)



For the most part, the average Chicagoan has walkable access to a bus stop.



No stops within half a mile
(centroids/tracts)

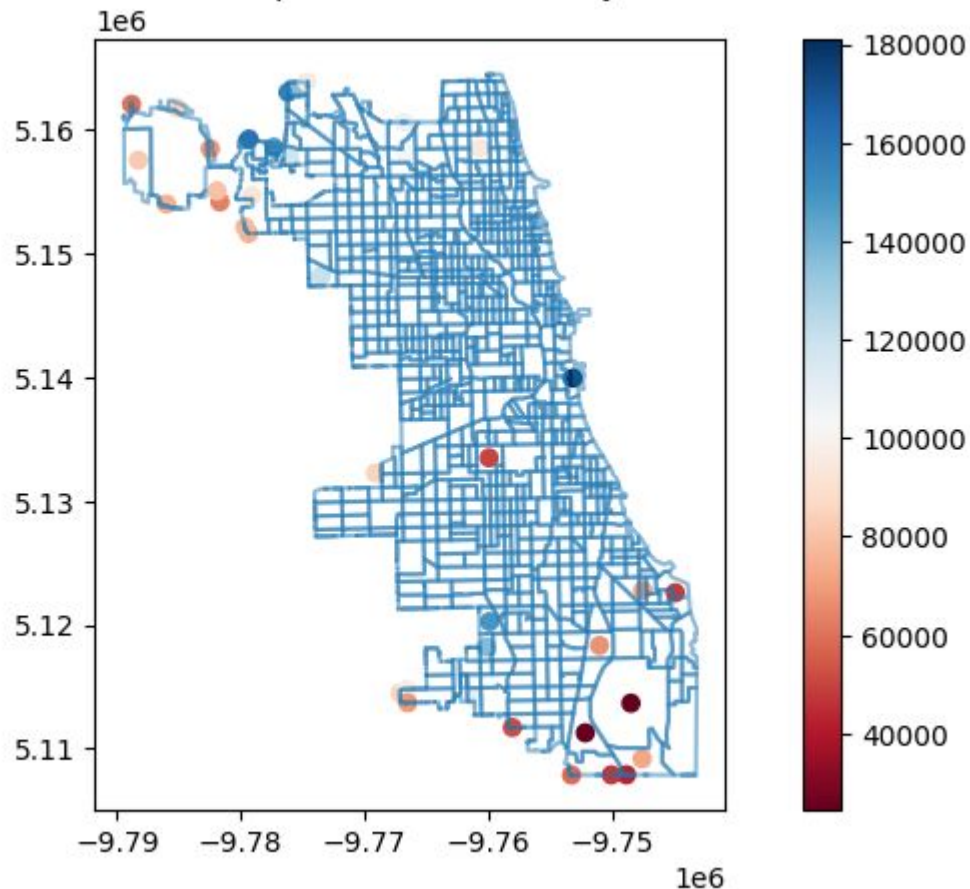
Census tracts without a bus stop within 0.5 miles by median household income

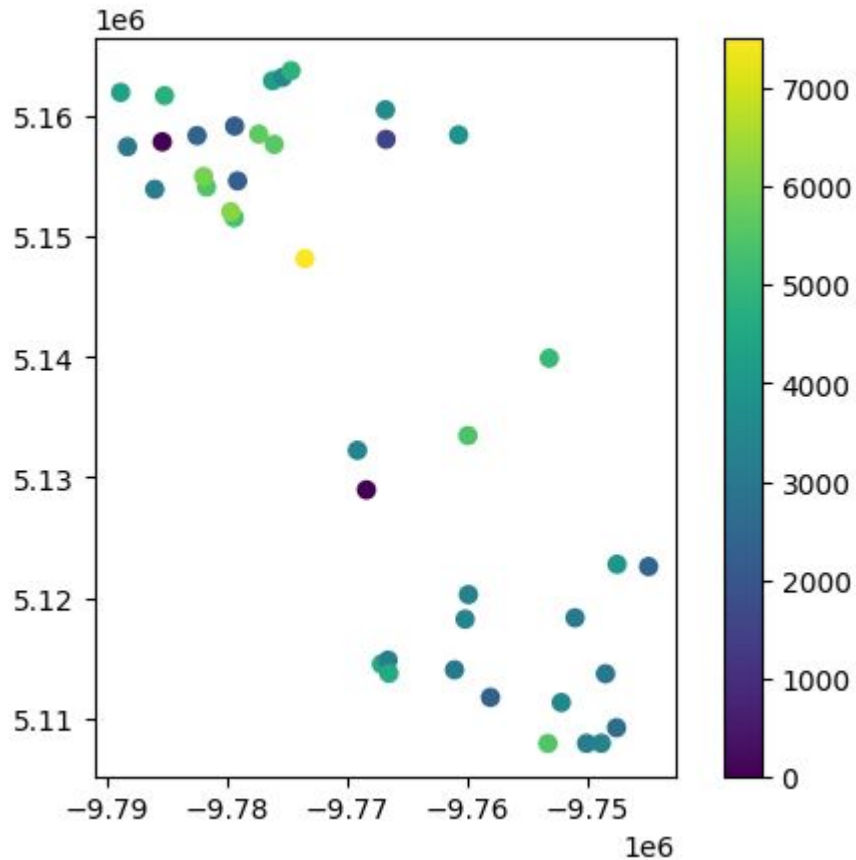
The census tracts in the south have:

- No L stop within 0.5 miles
- No bus stop within 0.5 miles
- Low income
- High disabled populations
- Commutes north of 40 minutes

Historically, the southern areas have:

- Been extensively redlined
- Lacked public development projects
- Received very little major business/jobs
- Felt the rust belt effect





154,428 people don't have walkable
access to a bus stop

1,792,859 people don't have walkable
access to an L stop (36%)

It doesn't have to be this way...

Population of census tracts without a bus stop within 0.5 miles

Conclusion

Chicago's public transport system is convenient for **millions**. Many cities in the US attempt to recreate Chicago's system. The city deserves it's credit as a pioneer in effective American transportation.

However

There are clear gaps in transportation in the southern and western neighborhoods. The maps I created showcase the reality of transportation justice in Chicago. The areas that contain transit deserts have a high disabled/foreign rate, low income, and long commute times.



GIS / Geopandas allow us to see injustices. In the future, I believe it would be immensely beneficial to collect data that documents the quality of each stop.